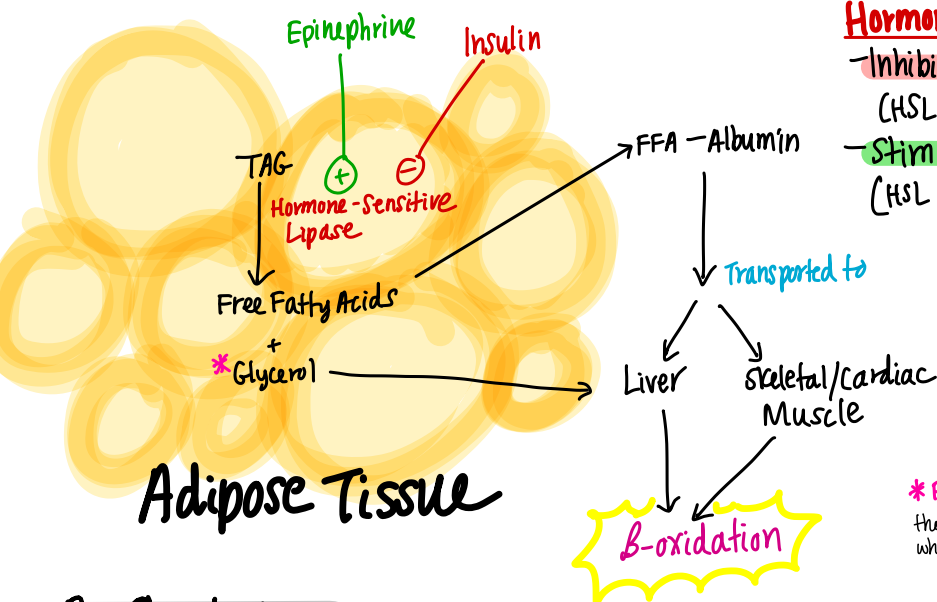


ADIPOSE TISSUE LIPOLYSIS AND BETA OXIDATION OF FATTY ACIDS

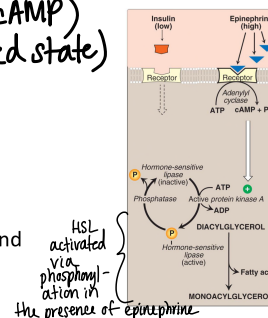


Hormone-sensitive Lipase

- Inhibited by insulin in the Fed state (HSL is inactive in dephosphorylated state)
- Stimulated by epinephrine (via cAMP) (HSL is active in the phosphorylated state)

Role of Epinephrine in fasting / flight / fight

The lipid droplets in the white adipose tissue are coated by a protein (perilipin)-phosphorylation by protein kinase A - facilitates phosphorylated HS lipase to bind to lipid droplet - stimulates hydrolysis of TAG to free fatty acids and glycerol



*** Fate of glycerol:** glycerol can't be used in adipose tissue because the adipose tissue lacks glycerokinase. Therefore, glycerol goes to the liver where it enters glycolysis or gluconeogenesis or triacylglycerol synthesis.

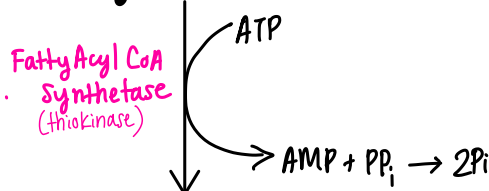
β -Oxidation

- oxidation of fatty acids at the β -carbon atom
- Intracellular location

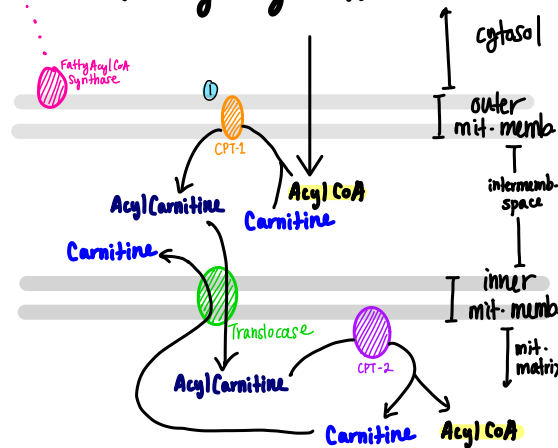
① Activation of fatty acid

- Fatty acid is activated to fatty acyl CoA via Fatty Acyl CoA synthetase (thiokinase)
- present in the outer mitochondrial membrane (cytosolic side)
- this step needs 2 high energy bonds (ATP broken down to AMP)

Fatty Acid + CoASH



Fatty Acyl CoA



② Transport of fatty acid from cytosol to the mitochondria

CPT = Carnitine palmitoyl transferase or carnitine acyl transferase

CPT-1 = outer mitochondrial membrane } different isoforms of CPT-1 & CPT-2 are present in the liver + muscle

CPT-2 = inner mitochondrial membrane

CPT-1 links acyl group to carnitine, which is the transporter. $\rightarrow$ becomes acylcarnitine. Acylcarnitine is transported through the inner mitochondrial membrane via translocase. CPT-2 then converts acylcarnitine back into acyl CoA + carnitine. The Acyl CoA continues onto β -oxidation while carnitine formed in mitochondrial matrix goes back into the intermembrane space via translocase.

Regulation: CPT-1 is inhibited by Malonyl CoA, which is an intermediate of fatty acid synthesis. Therefore when fatty acid synthesis is active, β -oxidation is inhibited at the level of CPT-1.

FYI: Fatty acids < 12 carbons can cross mit. memb. w/out CPT or carnitine.

③ β -oxidation proper

- sequence of 4 reactions involving β -carbon atom of fatty acid

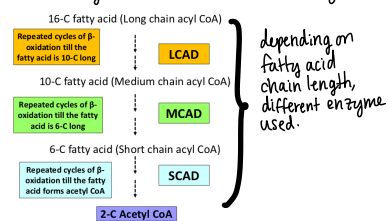
- ① Acyl CoA dehydrogenase (requires FAD) - oxidation
- ② Enoyl CoA hydratase - addition of water
- ③ 3-hydroxy acyl CoA dehydrogenase (requires NAD⁺) - oxidation
- ④ Thiolase - cleavage

Each cycle of β -oxidation forms ① acetyl CoA, ② FADH₂, and ③ NADH + H⁺

Complete β -oxidation of 16C palmitic acid results in the formation of

8 Acetyl CoA, 7 FADH₂, 7 NADH + H⁺

Each Acetyl CoA enters the Krebs cycle to form 12 ATP

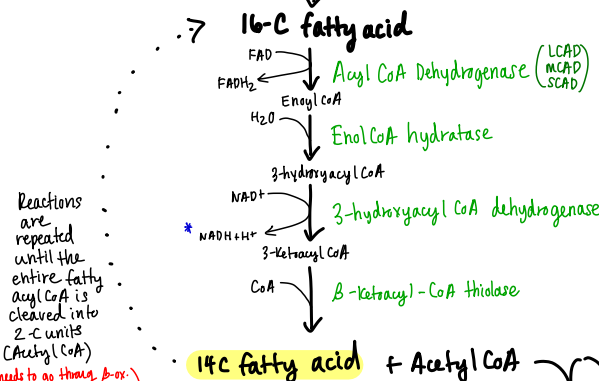


Energy yield from β -oxidation of palmitic acid

8 Acetyl CoA, 7 FADH₂, 7 NADH + H⁺

$$\begin{aligned} & \frac{8 \times 12 \text{ ATP (Krebs)}}{96 \text{ ATP}} + \frac{7 \times 2 \text{ ATP}}{14 \text{ ATP}} + \frac{7 \times 3 \text{ ATP}}{21 \text{ ATP}} = 131 \text{ ATP} \\ & 131 \text{ ATP} - 2 \text{ ATP (used to activate fatty acid)} = \text{net } 129 \text{ ATP} \end{aligned}$$

- for comparison, oxidation of glucose = 38 ATP



Reactions are repeated until the entire fatty acyl CoA is cleaved into 2-C units (Acetyl CoA)

(needs to go through β -ox. 7x to finally form 8 molecules of Acetyl CoA)

* NADH $\rightarrow$ needed for ketogenesis

Krebs

Disorders of Fatty Acid Oxidation

Common features of systemic fatty acid oxidation disorders

Hypoglycemia: cells increasingly rely on glucose for energy and there is reduced gluconeogenesis) - Acetyl CoA = absolute activator for pyruvate carboxylase in gluconeogenesis but not being produced b/c of fatty acid oxidation disorder.

Hypoketosis: due to reduced formation of Acetyl CoA

MCAD deficiency: most common inherited AR enzyme deficiency

- ↓
Systemic
- presents 6-24 mos of age
 - decreased ability to oxidize fatty acids 6-10C in length (med. chain FAs)
 - **severe hypoglycemia** in response to fasting or illness that reduces appetite.
 - These individuals are unable to utilize fatty acids as an energy source.
 - tissues can only rely on glucose for their source of energy resulting in the profound hypoglycemia.
 - long chain FAs are oxidized to medium chain Acyl CoAs (8-10 carbons)
 - medium chain acyl carnitines are excreted in urine - typical to see in urine + serum
 - dicarboxylic acids found in urine (due to ↑ed flux through ω-oxidation)
- LABS:** ↑FFA in blood but ↓ketone bodies

Consequences of MCAD deficiency

Decreased β-oxidation	• Decreased β oxidation of medium chain fatty acids • C8-C10 acyl carnitines in blood + ur • Increased flux through ω-oxidation (Dicarboxylic acids in urine)
Hypoglycemia	• Decreased utilization of fatty acids by peripheral tissues • Increased reliance on glucose as an energy source • Decreased ATP and acetyl CoA to activate gluconeogenesis
Hypoketonemia	• Decreased β-oxidation in liver • Decreased substrate for ketogenesis (acetyl CoA)

Systemic FA oxidation disorders

- **organs affected:** liver + muscle
 - **sys:** Hypoglycemia + hypoketosis
 - **caused by:**
 - MCAD deficiency
 - CPT-I deficiency
 - systemic carnitine def.
 - Jamaican vomiting sickness
- Manifests:** 6-24 mos

Myopathic FA oxidation disorders

- **organs affected:** Muscle (skeletal + cardiac)
- **sys:** Muscle cramps during exercise (esp. during aerobic/sustained exercise)
- **Muscle biopsy:** will show elevated TAG (lipid droplets) in the muscle
- **caused by:** Myopathic carnitine deficiency
 - CPT-II deficiency
- **manifests:** adolescence

Carnitine Deficiency

- carnitine uptake into tissues is impaired
- Transport of long-chain FAs into the mitochondria is impaired + β-oxidation is ↓ed.
- **systemic** carnitine deficiency presents @ an early age
 - hypoglycemia due to impaired gluconeogenesis (↓ Acetyl CoA to activate pyruvate carboxylase)
 - ketogenesis is ↓ed if liver carnitine is deficient
- **myopathic** carnitine deficiency presents @ later age
 - muscle weakness + cardiomyopathy
 - presence of CK-MM & myoglobin in urine = skeletal muscle damage
 - lipid droplets in muscle biopsy

CPT-I + CPT-II deficiency

- **CPT-I** deficiency = hypoglycemia + hypoketosis; commonly affects the liver isoform (systemic form)
- **CPT-II** deficiency = cardiomyopathy + muscle weakness (myopathic form)
 - lipid deposits (TAGs) in skeletal muscle
 - prolonged exercise results in myoglobinuria + ↑ CK-MM levels in serum
 - CPT-II def. → commonly affects muscle

Jamaican Vomiting Sickness

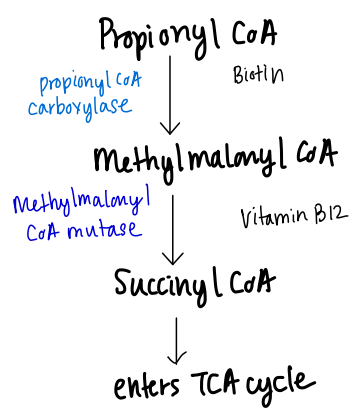
- hypoglycin from unripe Ackee fruit
- Hypoglycin A inhibits MCAD
 - ↳ ↓ β-oxidation
 - ↳ ↓ Acetyl CoA, ∴ ↓ gluconeogenesis
- Onset of vomiting 2-6 hrs s/p ingestion
- medium-chain acyl carnitines found in urine
- drowsiness, coma, & death if not immediately recognized + treated

Management of FA oxidation disorders

- **Systemic Fatty acid oxidation disorders** - MCAD deficiency/ CPT-I deficiency/ Carnitine deficiency (primarily management of hypoglycemia)
 - Management of **hypoglycemia** by using **IV glucose**
 - **Frequent feeding and high carbohydrate, low fat diet. Avoid fasting**
- **Myopathic fatty acid oxidation disorders** - Myopathic carnitine deficiency/ CPT-II deficiency (symptoms are primarily myopathic)
 - **Cease muscle activity and give glucose**
 - **Carnitine supplementation**

Oxidation of odd-chain FAs

- oxidation of odd-chained FAs produces propionyl CoA (3C) in the final round of β-oxidation
- propionyl CoA is converted to succinyl CoA (via propionyl CoA carboxylase and methylmalonyl CoA mutase)
- succinyl CoA enters the TCA cycle



Regulation of β-oxidation

- Serum free fatty acid levels primarily regulate the rate of fatty acid oxidation
 - Serum free fatty acid levels are regulated by insulin/ glucagon ratio. Low insulin/ glucagon ratio (during fasting) activates **hormone sensitive lipase** in adipose tissue (**lipolysis**)
- Entry of fatty acids into the mitochondria is regulated by CPT-I
 - **Malonyl CoA** (formed during fatty acid synthesis) inhibits CPT-I. This ensures that β-oxidation and fatty acid synthesis do not take place simultaneously

